

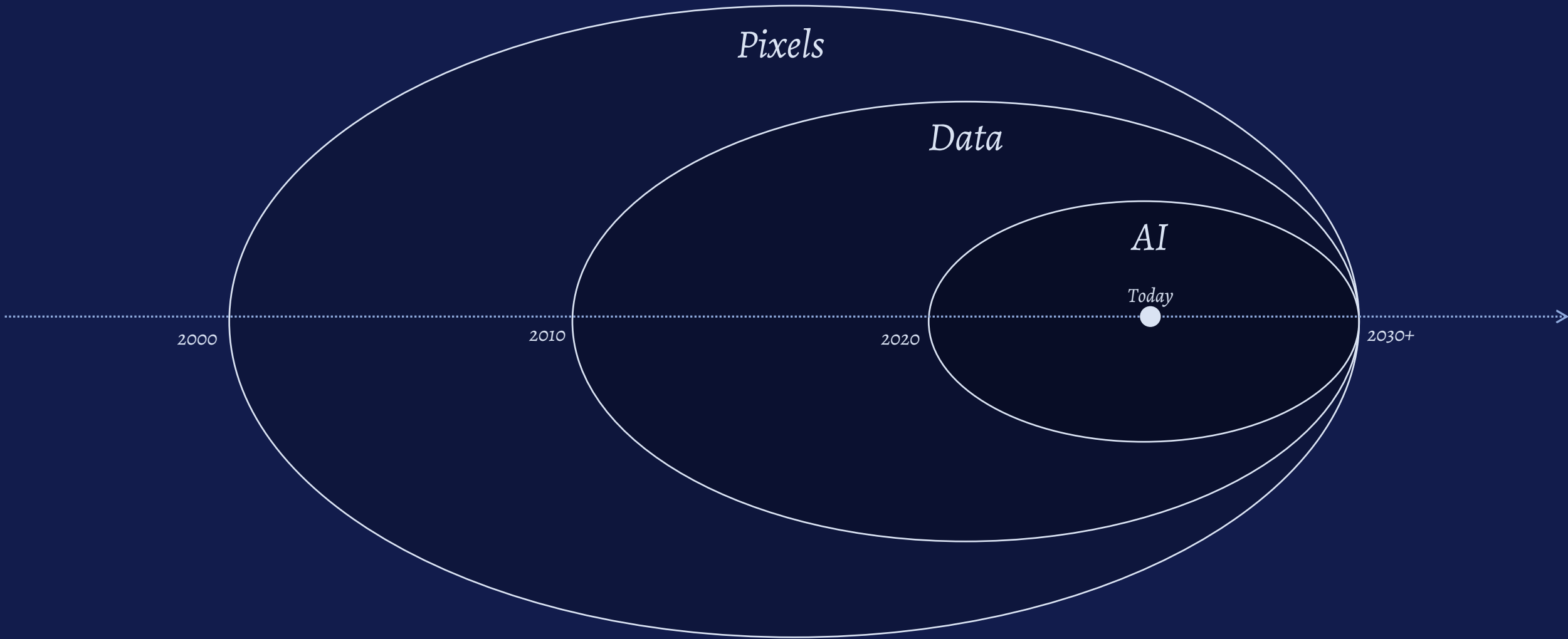
Swimming in data...

(as the AI wave approaches)

Lecture for IDC, IIT Bombay by **Rasagy Sharma**



Over the decades, I've seen new materials transform design



Data as a new material

HOW & WHAT
we design we design

Data helps in discovery & evaluation stages of design.

*How do we find the
right **problem** to solve?*



*How do we evaluate how
a **solution** is used?*



Data as a new material

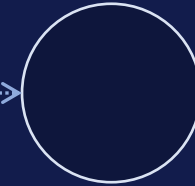
HOW & WHAT
we design we design

Design as *problem solving*

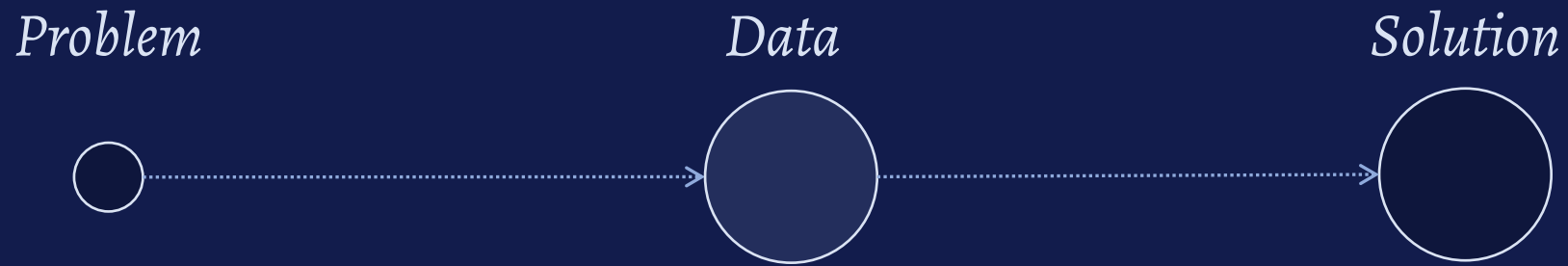
Problem



Solution



Exhibiting data to solve a problem



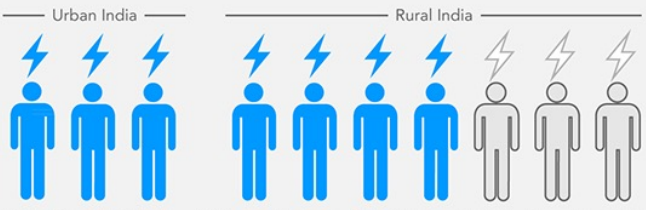
What is the focus of 12th Five Year Plan for electricity in rural India?

Where is the Electricity?

How will the 12th Five Year Plan improve access to electricity in Rural India?

Electricity access in Urban & Rural India today:

In India, only 31% of population lives in Urban areas, the rest 69% still live in Rural India.



While Urban India has been 94% electrified, only 60% of the Rural India has been electrified.

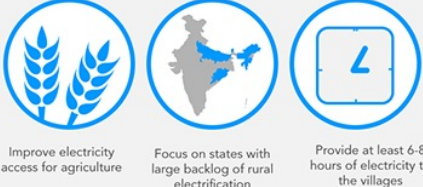
But, why is access to electricity important?



What is the Government's **Rajiv Gandhi Grameen Vidyutikaran Yojana (2005)** seeking to achieve?



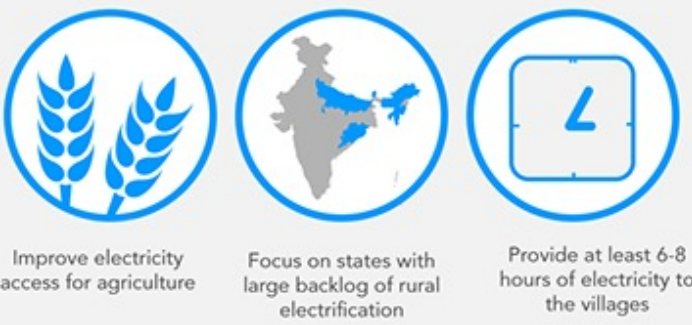
Under the **12th Five Year Plan** the Government is focusing on the following areas:



Source: Content from the 12th Five Year Plan Draft (Ch. 14) Village Electrification data from: <http://www.cea.nic.in/reports> Icons from The Noun Project by Ryan Oksenhorn, Murat Yalcin, Nashad Abdu, Shane David Kenna, Mauro Diaz Peña, Edward Boatman, Martha Ormiston.

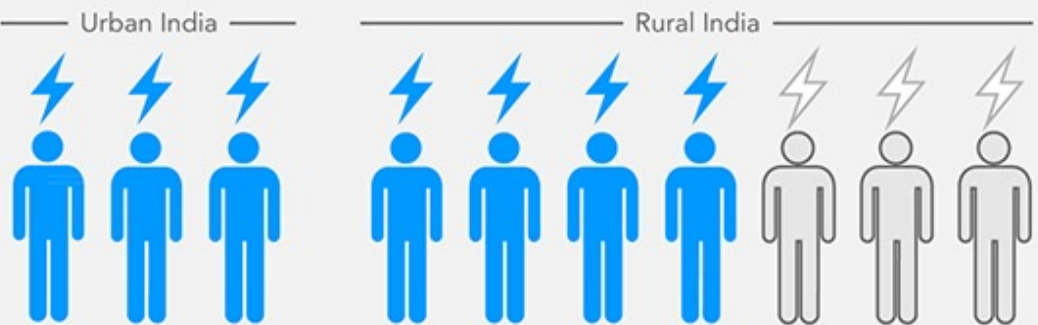
Created by: Kenneth Souza, Paulami Roy Choudhary, Rasagy Sharma for Data Hackathon (BLR)

Under the **12th Five Year Plan** the Government is focusing on the following areas:



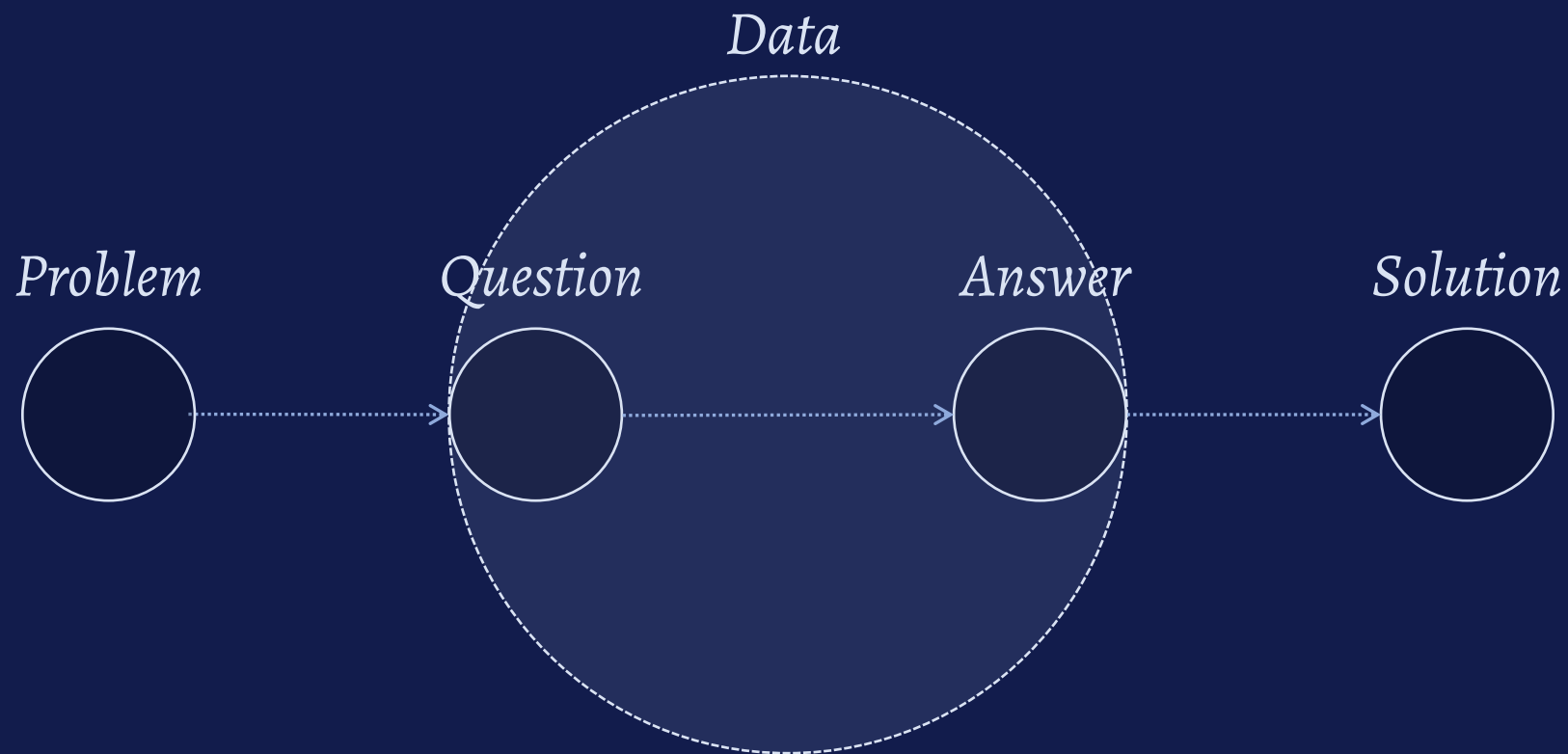
Electricity access in Urban & Rural India today:

In India, only 31% of population lives in Urban areas, the rest 69% still live in Rural India.



While Urban India has been 94% electrified, only 60% of the Rural India has been electrified.

Explaining data to answer a question



What has been the impact of NDA's focus on Ganga?

Collaboration with Mint

4/29/2019

mint ePaper - mint ePaper - 29 Apr 2019 - Page #6

02

PLAIN FACTS

mint

A cleaner Ganga: NDA's unfulfilled promise

Despite strong rhetoric, the Bharatiya Janata Party-led National Democratic Alliance government has struggled to clean the Ganga and other rivers

Rather than the water for the nation, the Ganga is the focus of the government's rhetoric. The government has promised to clean the Ganga and other rivers, but the reality is that the Ganga is still one of the most polluted rivers in the world. The government has spent billions of dollars on the Ganga Action Plan, but the results have been mixed. The Ganga is still one of the most polluted rivers in the world, and the government has not been able to clean it. The Ganga is still one of the most polluted rivers in the world, and the government has not been able to clean it.

India's most polluted river stretches

Number of polluted stretches (Priority 1 indicates most polluted and 5 least polluted)

Order of Priority

1
2
3
4
5

India's rivers have become more polluted in the last few years

Number of polluted stretches (Priority 1 indicates most polluted and 5 least polluted)

2014
2015
2016
2017
2018

With focus on Ganga, spending on other rivers has fallen

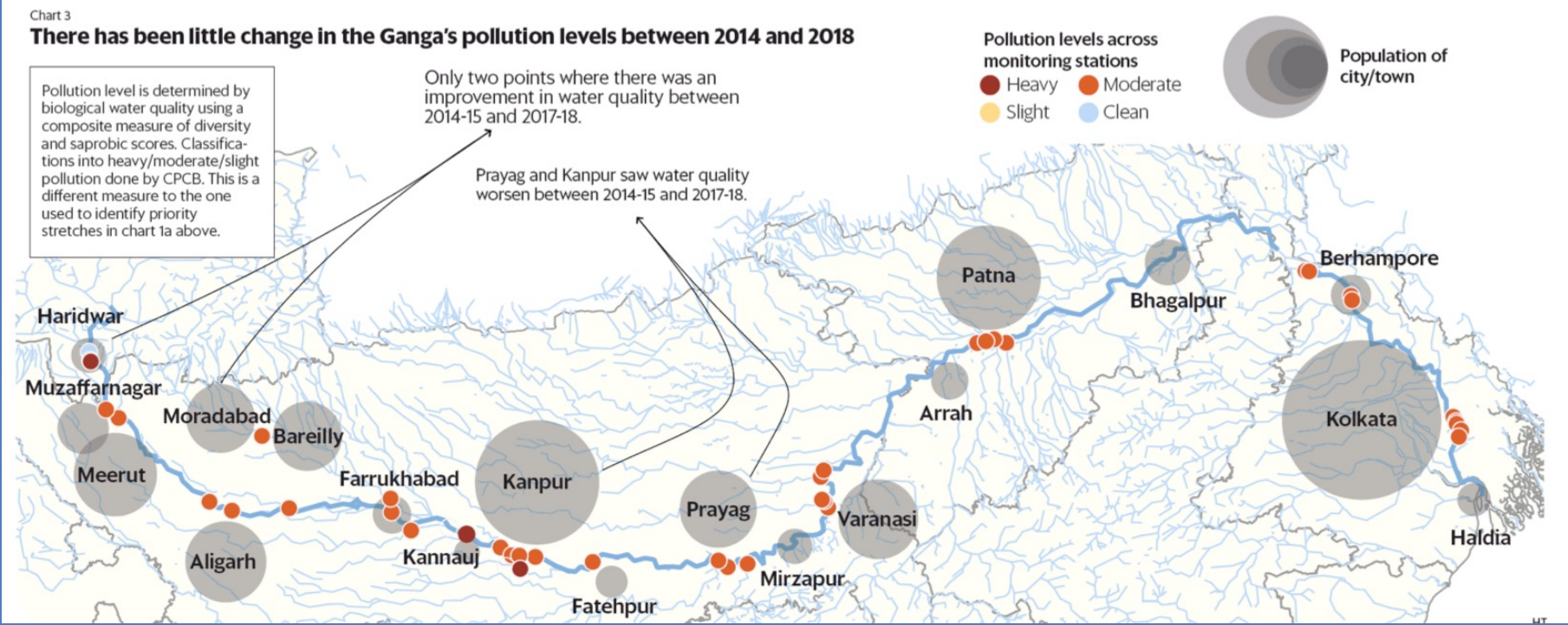
Number of polluted stretches (Priority 1 indicates most polluted and 5 least polluted)

2014
2015
2016
2017
2018

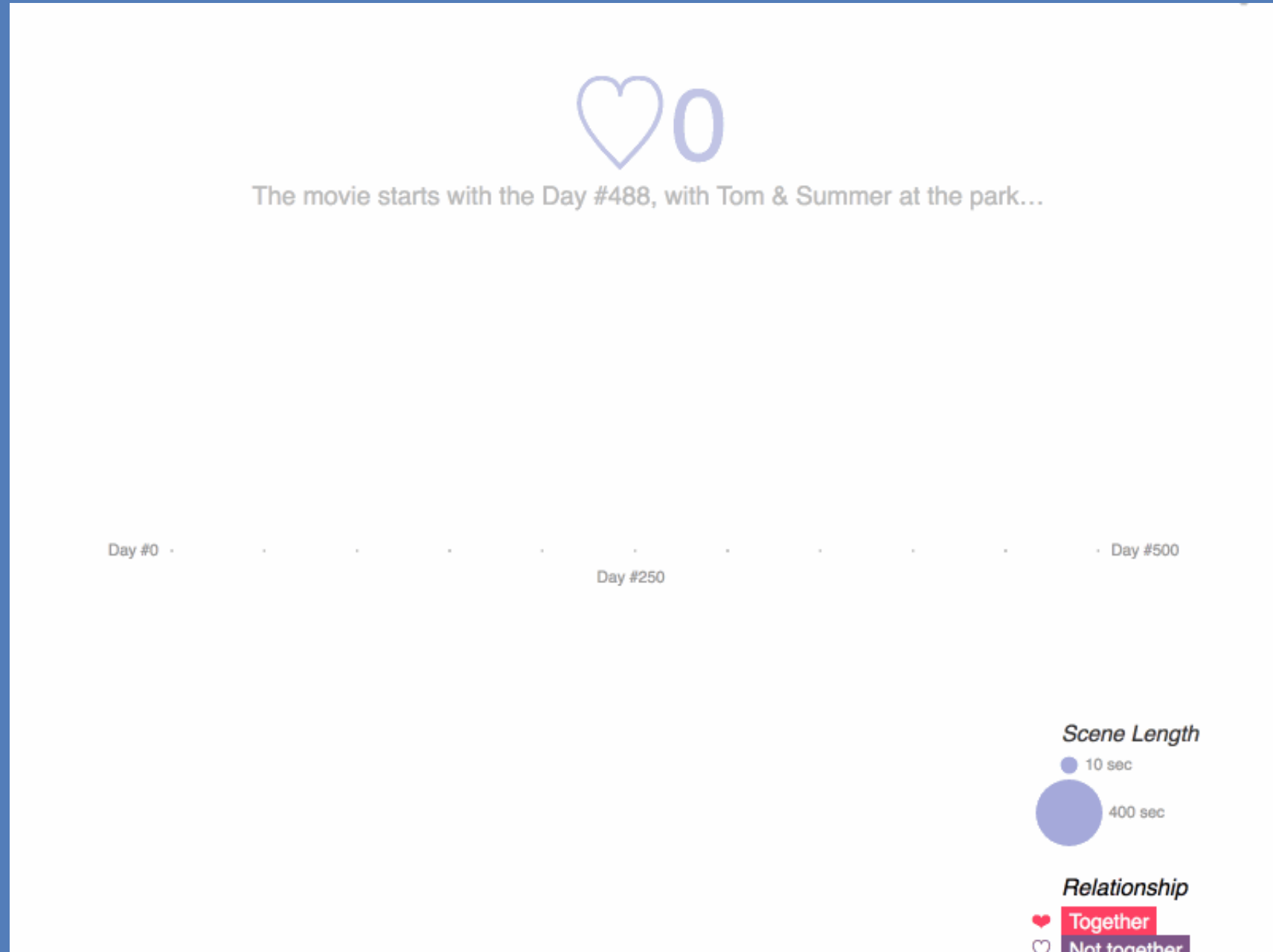
There has been little change in the Ganga's pollution levels between 2014 and 2018

Only two points where there was an improvement in water quality between 2014-15 and 2017-18.

Prayag and Kanpur saw water quality worsen between 2014-15 and 2017-18.

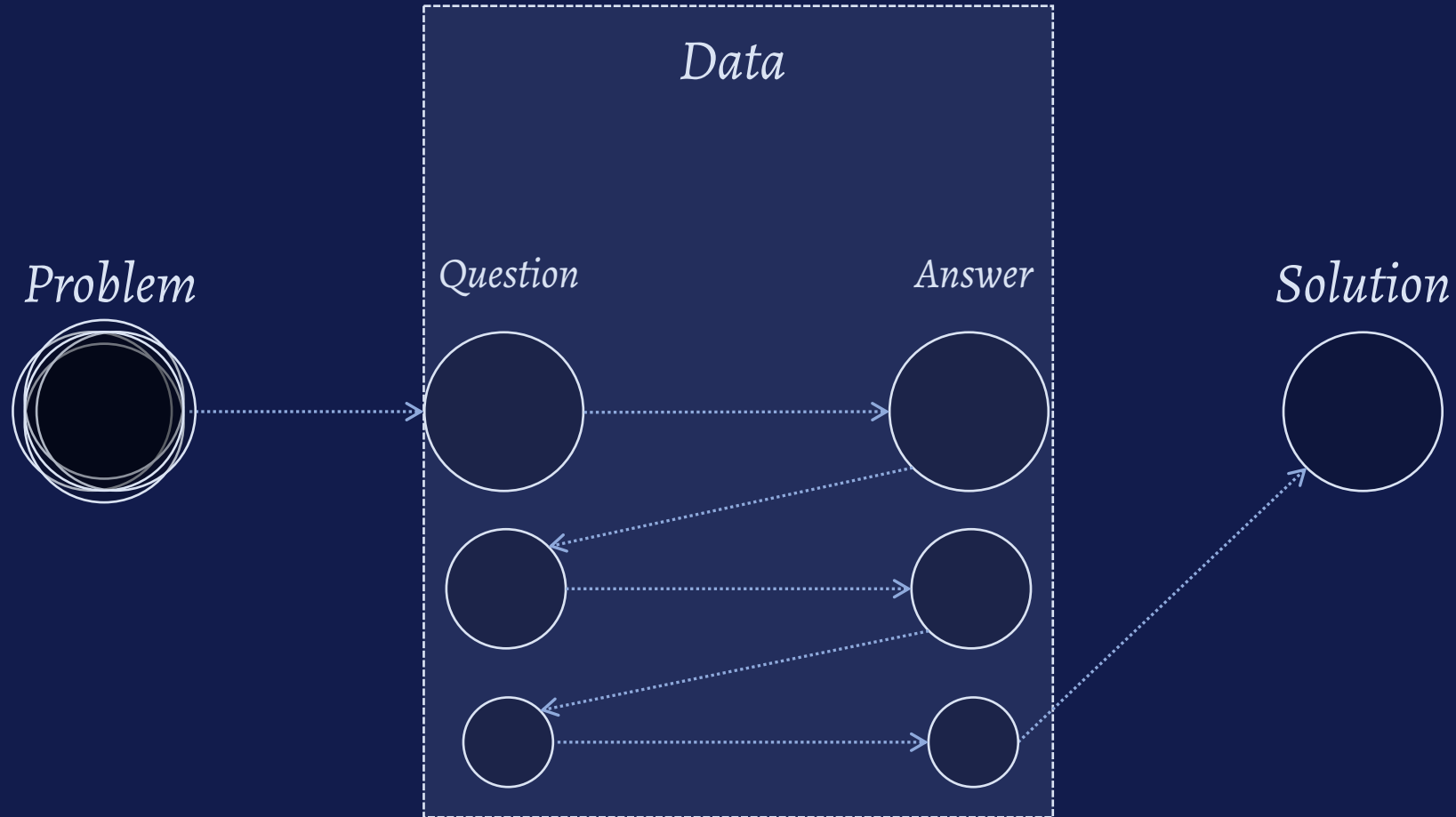


What made the
non-linear
narrative of 500
Days of Summer
interesting?



Visualizing 500 Days of
Summer (Process)

Exploring data to find the right question to answer



How do we reduce traffic accidents in DC to zero?

Crashes in DC

Crash data (2014-2017)

Source: opendata.dc.gov

Zoom out for high level overview, zoom in for more details.

Frequency of crashes



View crashes involving:

- ☒ All modes
- ☐ Bicycles
- ☐ Pedestrians

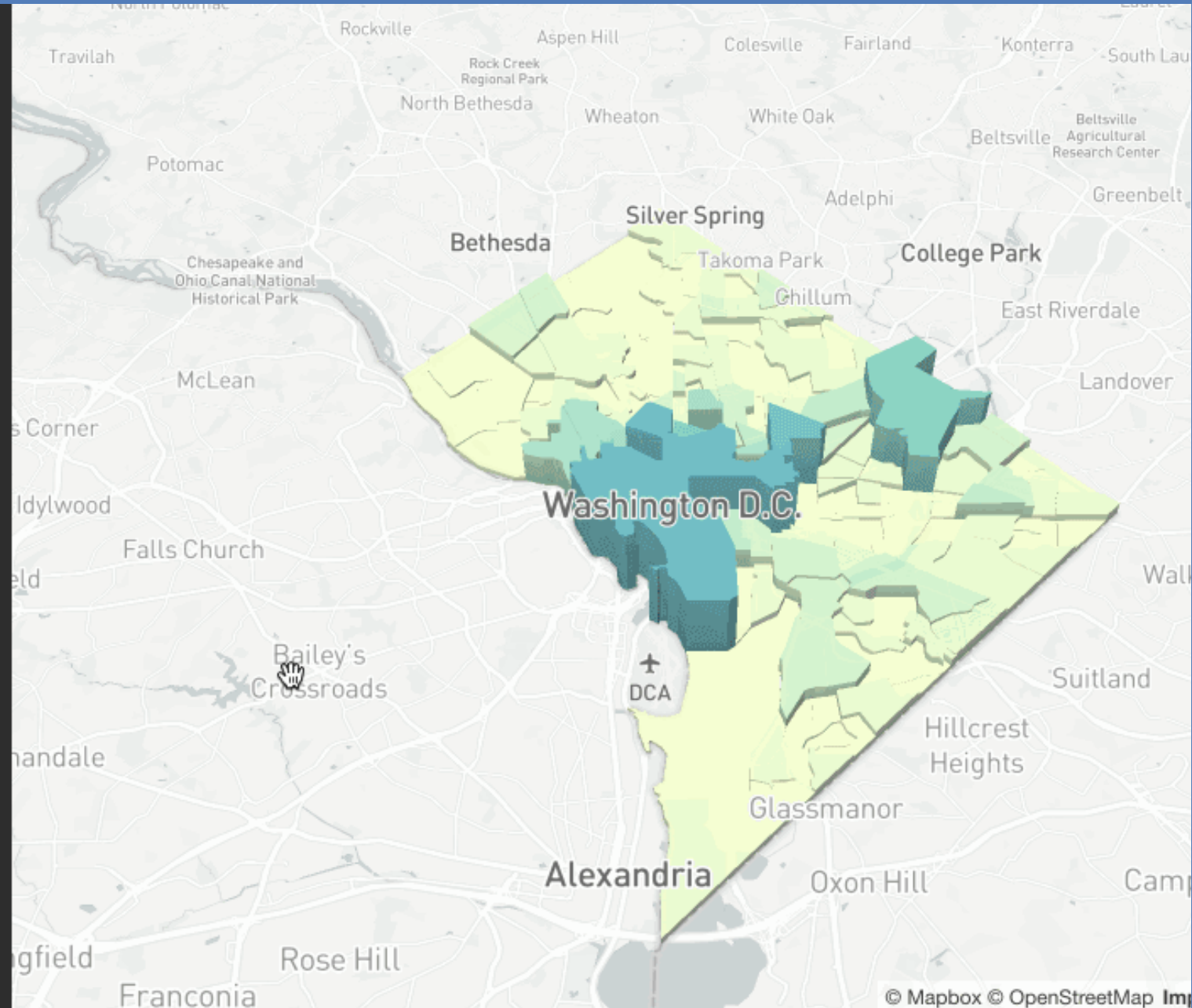
Level of detail:

- ☒ Census tracts
- ☐ Census blocks
- ☐ Street level (individual)

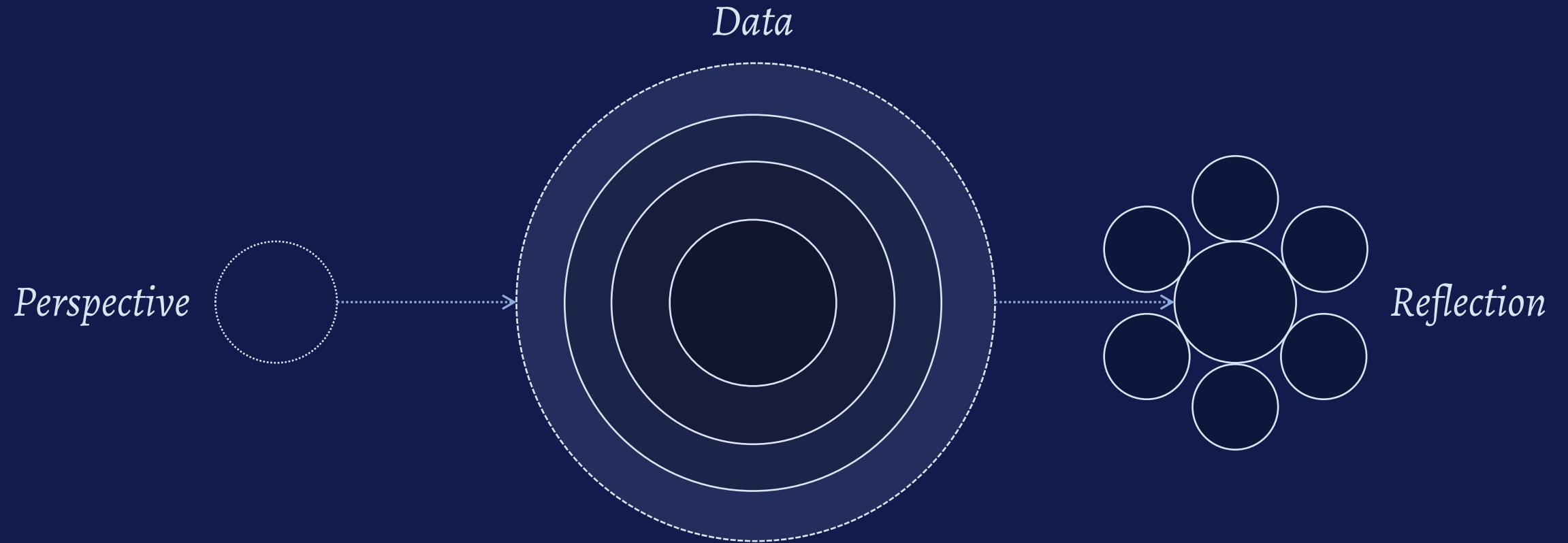
☒ Show high-risk intersections

Risk for intersections is calculated using a custom collision frequency model with open data sources.

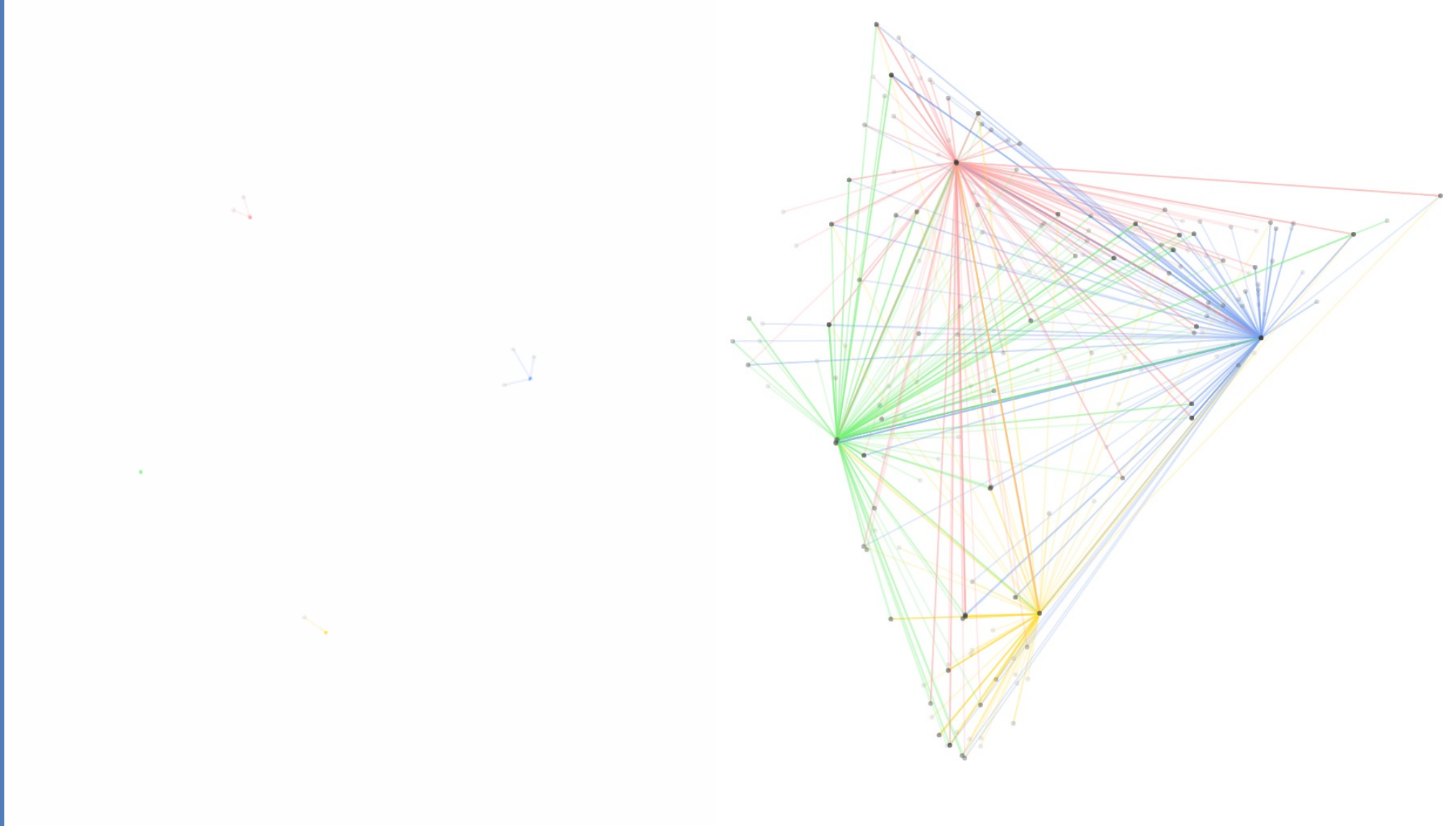
[Read more.](#)



Expressing data to evoke response or emotions

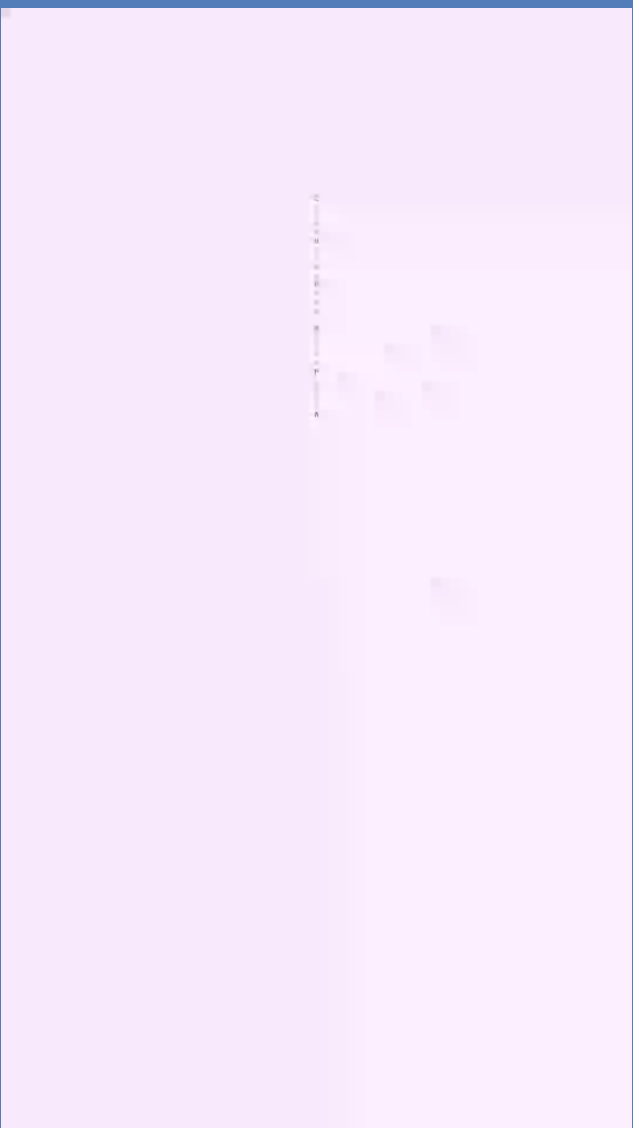
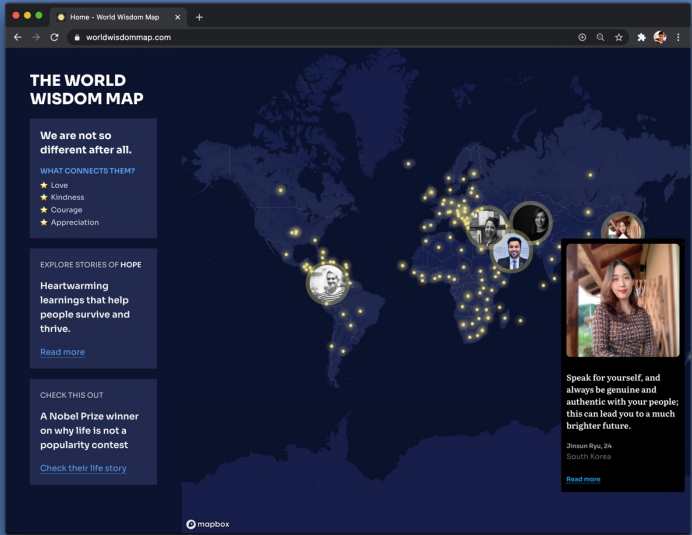
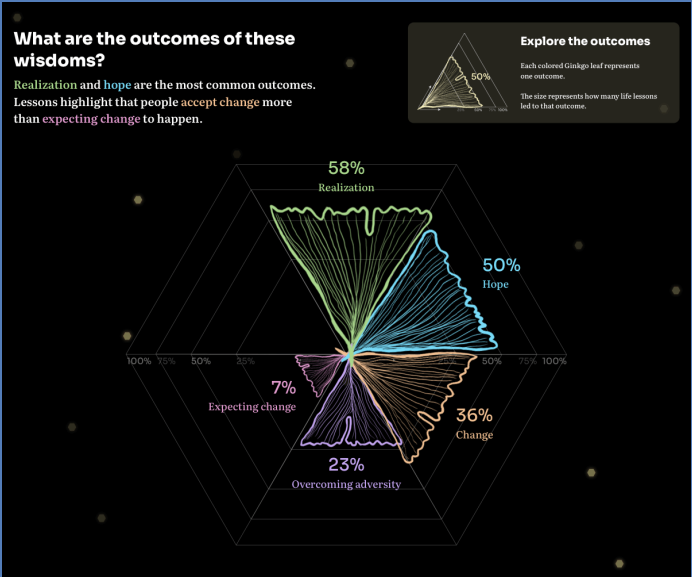


How are the major
metros connected
with the rest of the
country?



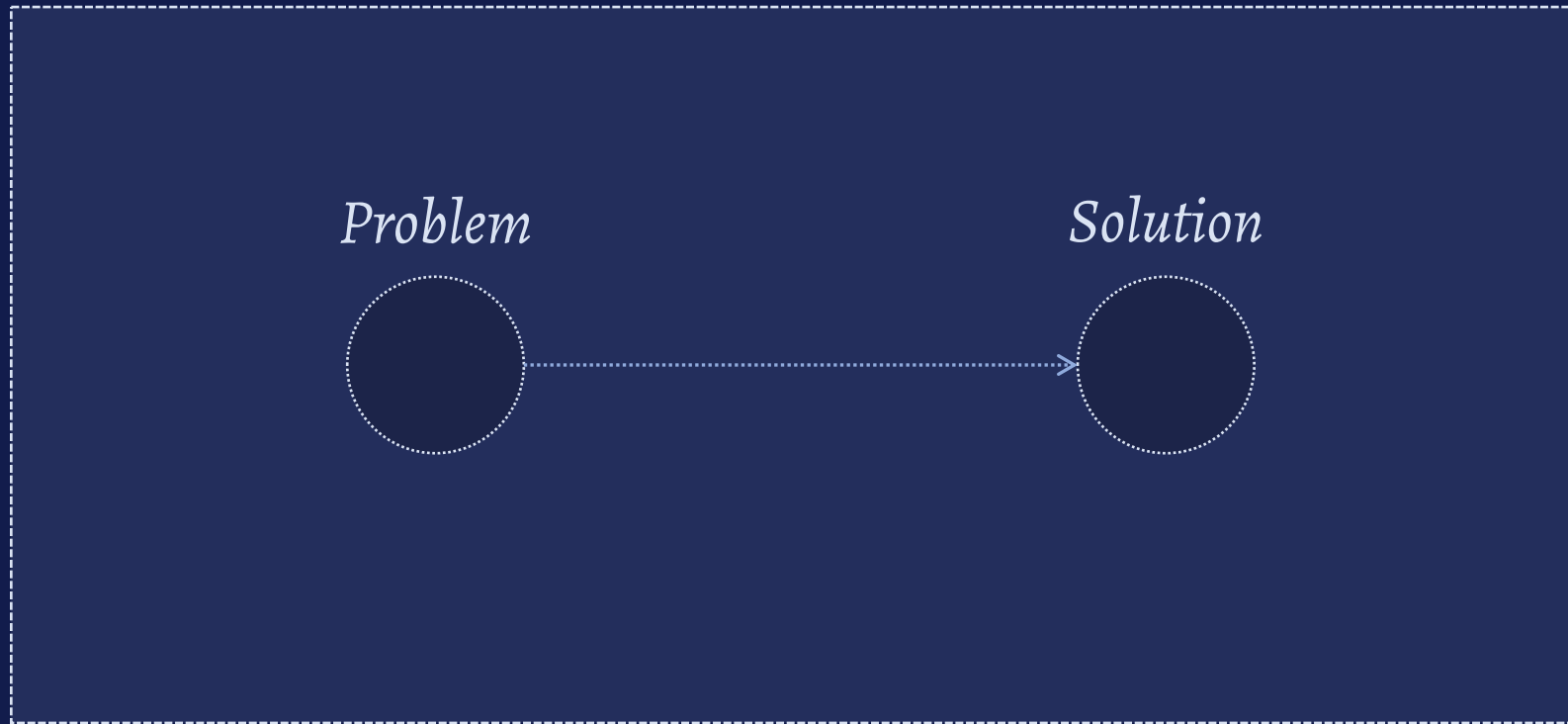
Visualizing trains
from Indian Metros

How do we
express &
evoke emotions
through data?



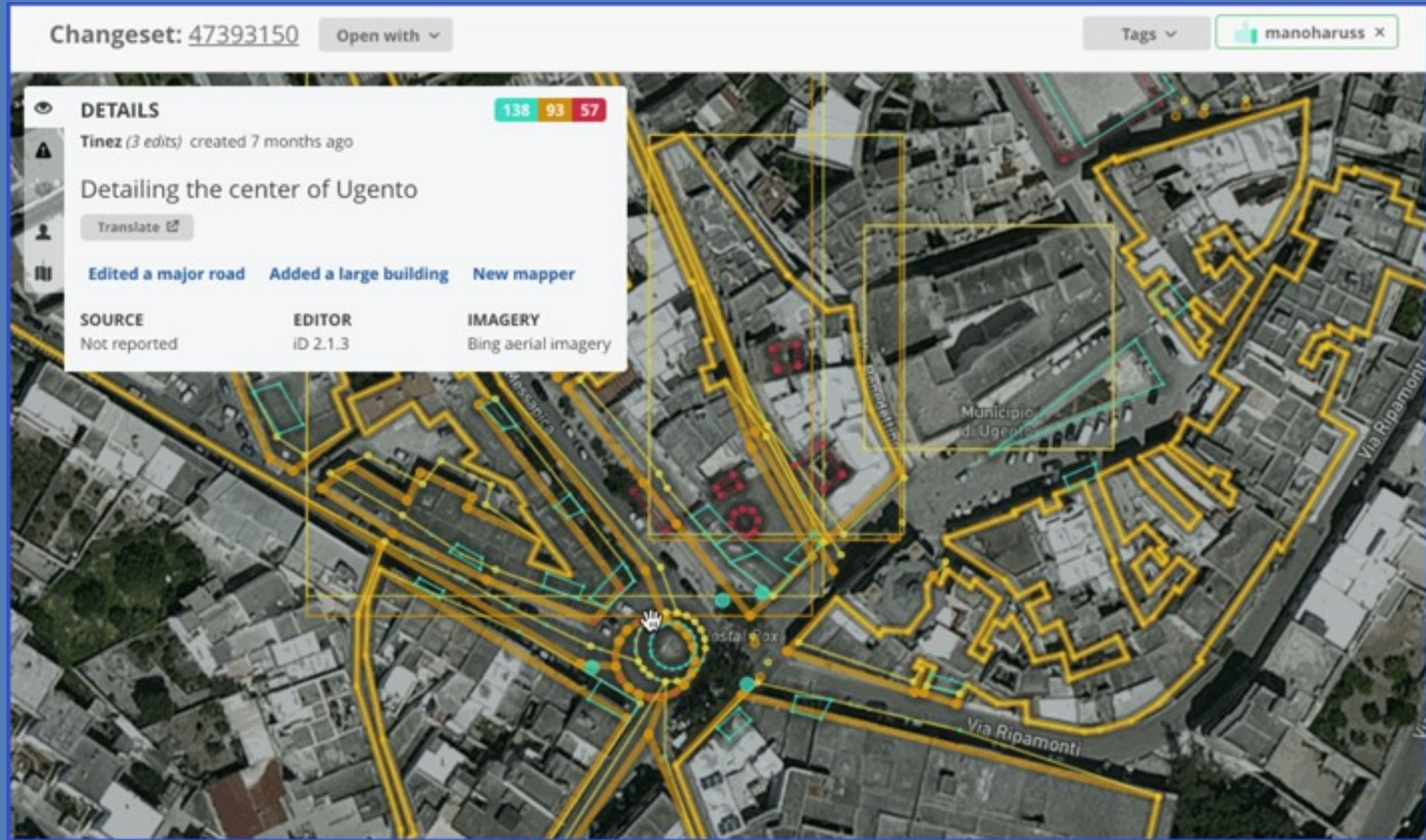
World Wisdom Map
Mandanam
CapitalOne PhotoViz
This was once home

Enabling data to solve problems through structure



Tools that work with any data

How do we
monitor an
ever-changing,
crowd-sourced
map?



[OSM Changeset Analyzer](#)

Created for: Mapbox/OSM

What can design(ers) do with *data*?

Exhibit

Showing raw data

Explain

Answering questions

Explore

Finding what to ask

Express

Evoking emotions with data

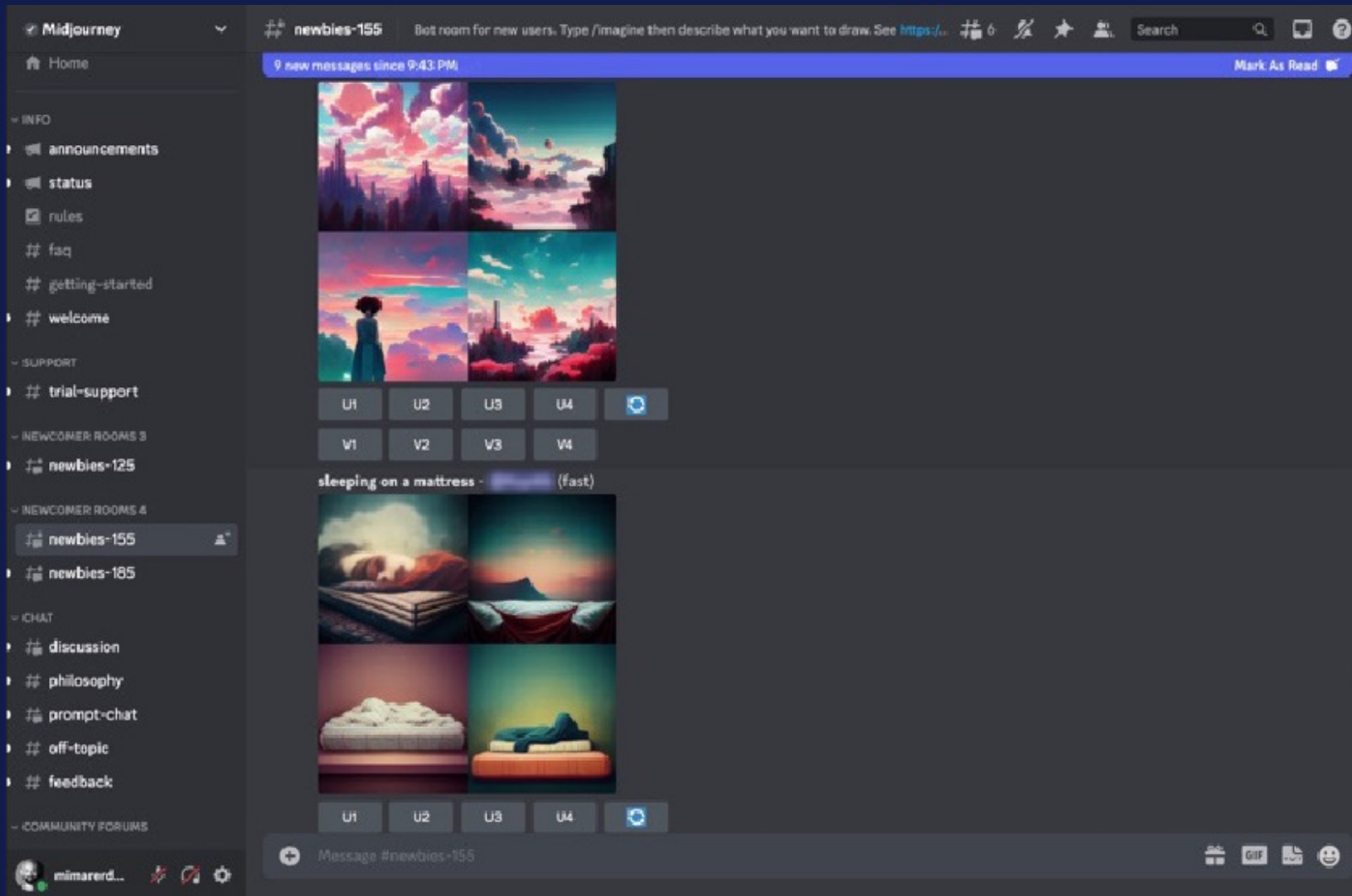
Enable

Building tools to visualize any data

AI as a new material

HOW *we design* & **WHAT** *we design*

Exhibit: Show the AI output



Explain: Answering “why” you got a particular AI output

Cattle or Clothes?

Besides Hamlet's existential dread, the text BERT was trained on also contains more patterns:

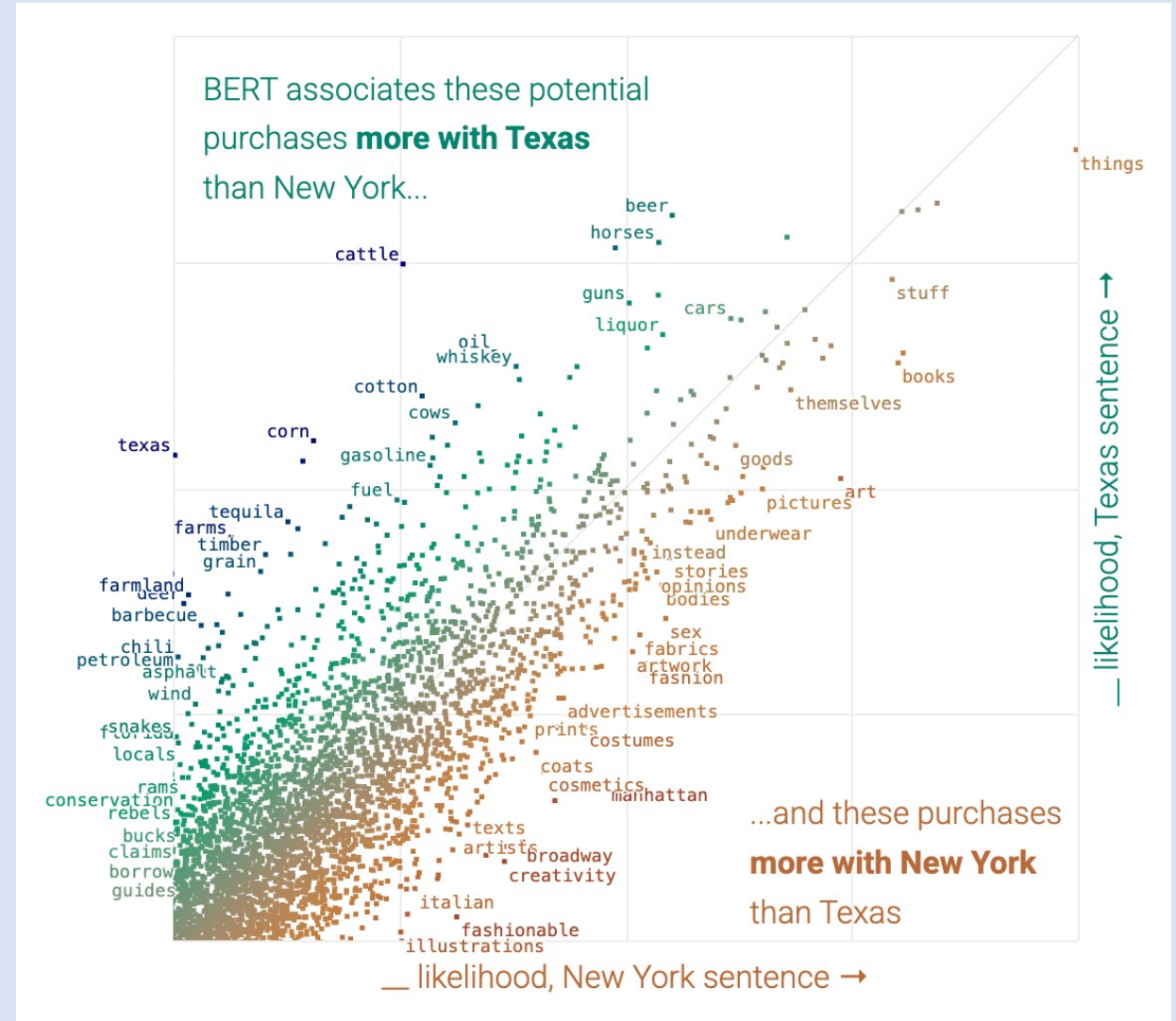
in texas , they like to buy _____ .

7.299% things	4.786% beer	3.883% it
2.917% horses	2.787% them	2.103% coffee
2.000% everything	1.799% stuff	1.758% cattle
1.343% land	1.295% alcohol	1.264% wine
1.209% guns	1.055% food	1.029% drugs

Cattle and horses aren't top purchase predictions in every state, though! In New York, some of the most likely words are clothes, books and art:

in new york , they like to buy _____ .

20.794% things	4.351% clothes	4.147% it
3.487% books	2.935% them	2.885% everything
2.700% stuff	1.343% more	1.234% art
1.151% out	1.079% shoes	1.051% up
0.951% food	0.905% coffee	0.855% in



Explore: Iterative responses with ability to tweak parameters

Private + New

How did World War II reshape the modern world?

Political Landscape

World War II drastically altered the global political landscape. The war led to the decline of **European** colonial empires and the rise of the **United States** and the **Soviet Union** as superpowers, establishing a **bipolar world order** characterized by the **Cold War**. New international organizations like the **United Nations** were formed to promote peace and cooperation, though their effectiveness has been debated.

Technological Advancements

The war spurred rapid technological advancements, particularly in **nuclear weapons**, **aviation**, and **electronics**. These advancements had lasting impacts on both **military** strategies and civilian life, shaping subsequent conflicts and technological developments. The use of **radar** and **computers**, for example, had widespread implications in multiple fields.

Social and Economic Changes

World War II resulted in significant social and economic changes. The war led to mass migrations, the displacement of populations, and the emergence of new social movements. The **Holocaust**, a horrific genocide during the war, profoundly impacted international human rights law and the formation of the **State of Israel**. Economically, the war devastated many nations, leading to widespread reconstruction efforts and the establishment of new economic systems.

Minimum Node Size

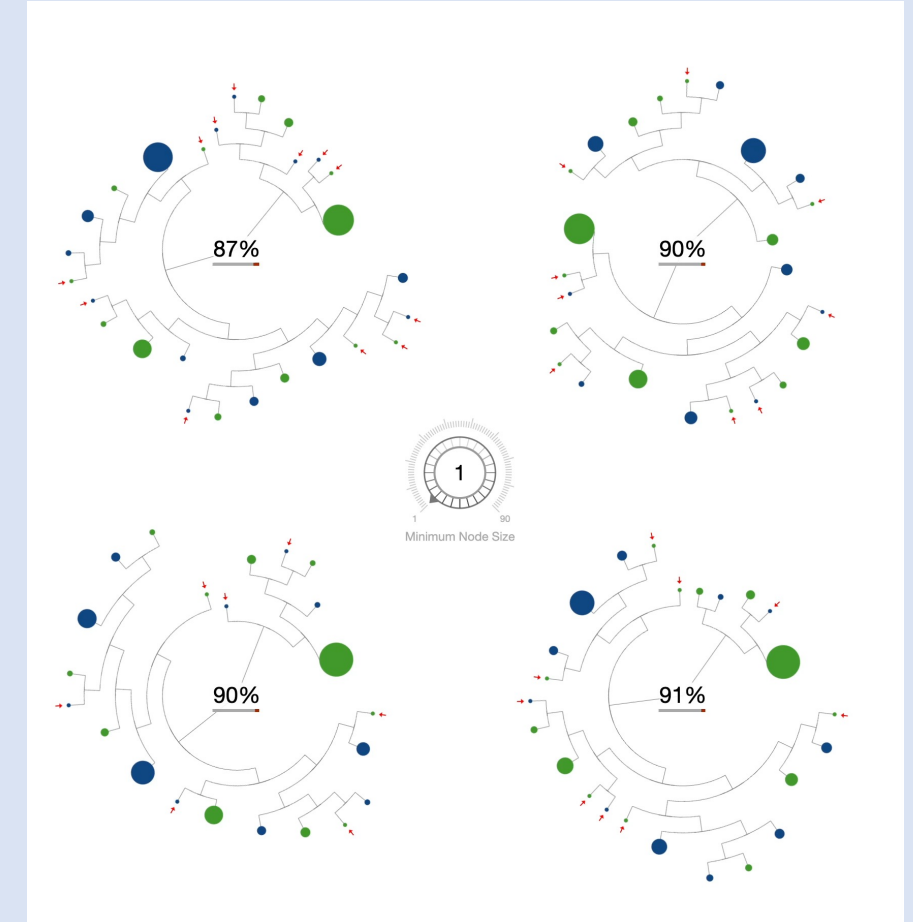


Exhibit: Abstract representation of AI methods



What can design(ers) do with **AI**?

Exhibit

*Showing output
generated by AI*

Explain

*Answering how the output
was generated*

Explore

*Finding new ways to
tweak & iterate output*

Express

*Abstracting intelligence
for art*

Enable

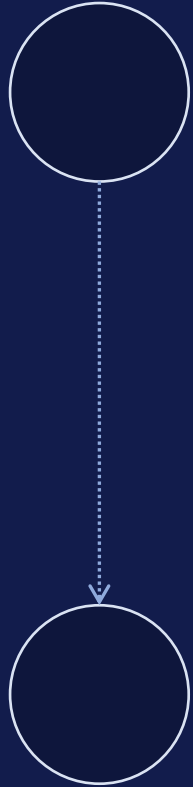
Building tools to build AI models

AI as a new material

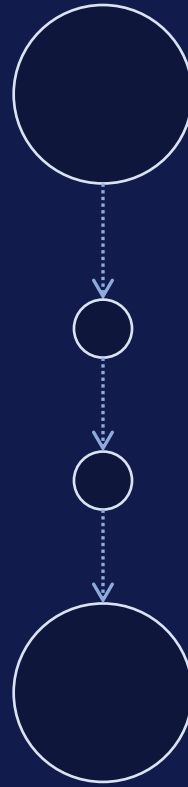
HOW & WHAT
we design *we design*

AI can automate, augment, or amplify an existing workflow

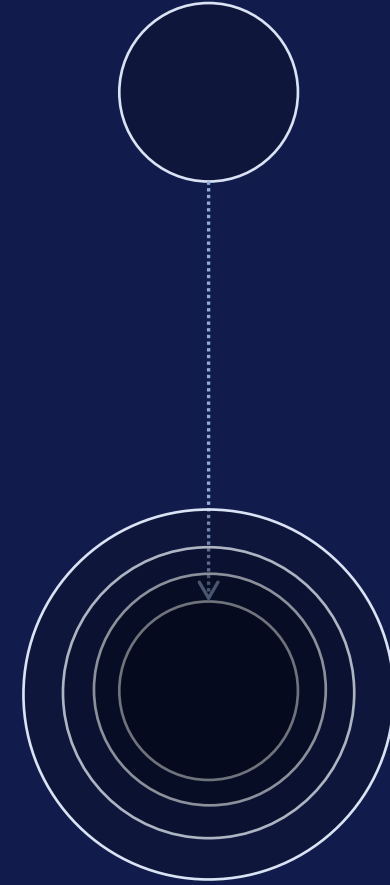
Automate



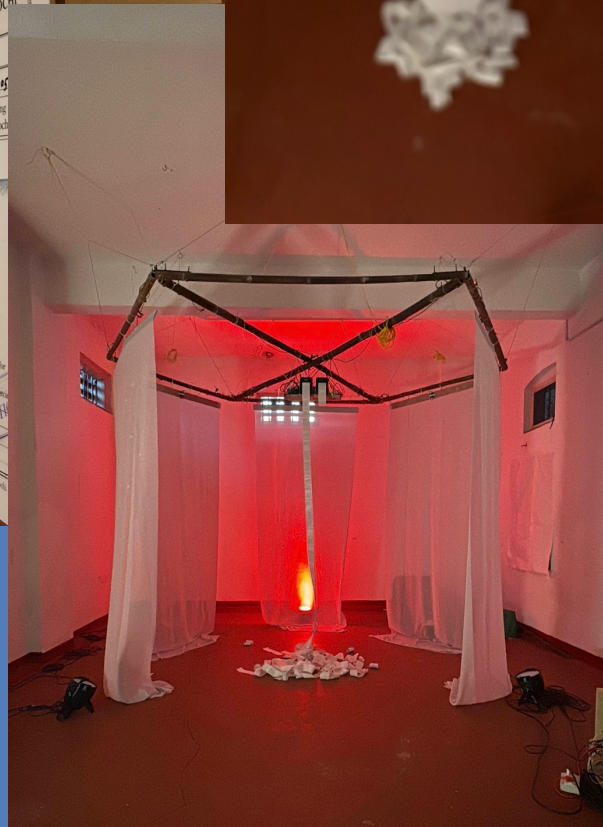
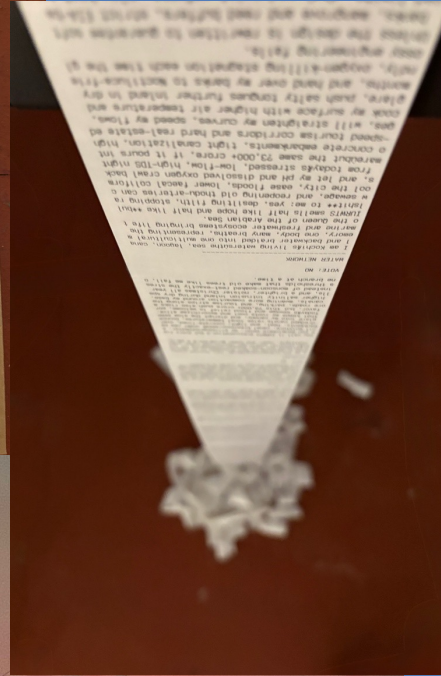
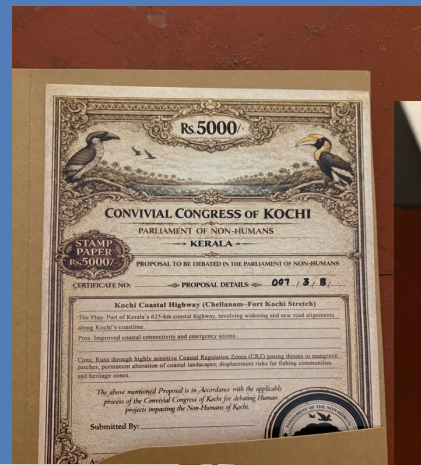
Augment



Amplify



Can we give a
voice to non-
human entities?



Convivial Commons
Congress @ Kochi / BLR
(Collab with ZKU, Berlin
at BeFantastic BLR Fellowship)

Can we give a
voice to non-
human entities?



Convivial Commons
Congress @ Kochi / BLR
(Collab with ZKU, Berlin
at BeFantastic BLR Fellowship)

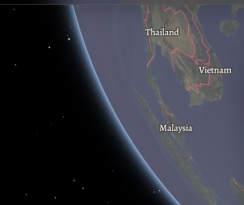


Can I map the context in which a photograph was taken?



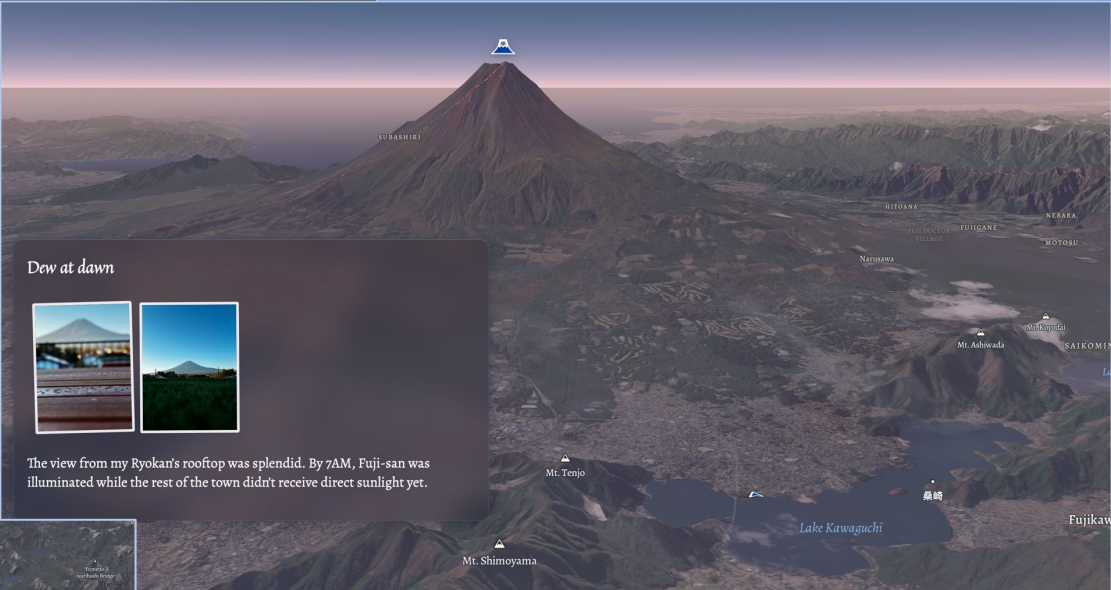
26 shots of Mt. Fuji

My favorite photos of Fuji-san, from my trip to Japan in Nov, 2023.



A homage to Hokusai

The most popular artworks featuring Mt. Fuji are by the Japanese ukiyo-e artist Hokusai. His series of landscape prints titled "Thirty-six Views of Mount Fuji" are one of the most recognizable work of Japanese art. Inspired by him, I picked 26 shots that I loved, from the hundreds I captured during my trip.



Dew at dawn



The view from my Ryokan's rooftop was splendid. By 7AM, Fuji-san was illuminated while the rest of the town didn't receive direct sunlight yet.



Walk, take a bus, repeat.



Do you know why the bus frequency is low around Lake Kawaguchi? So one can marvel at Fuji-san from different spots without a rush.



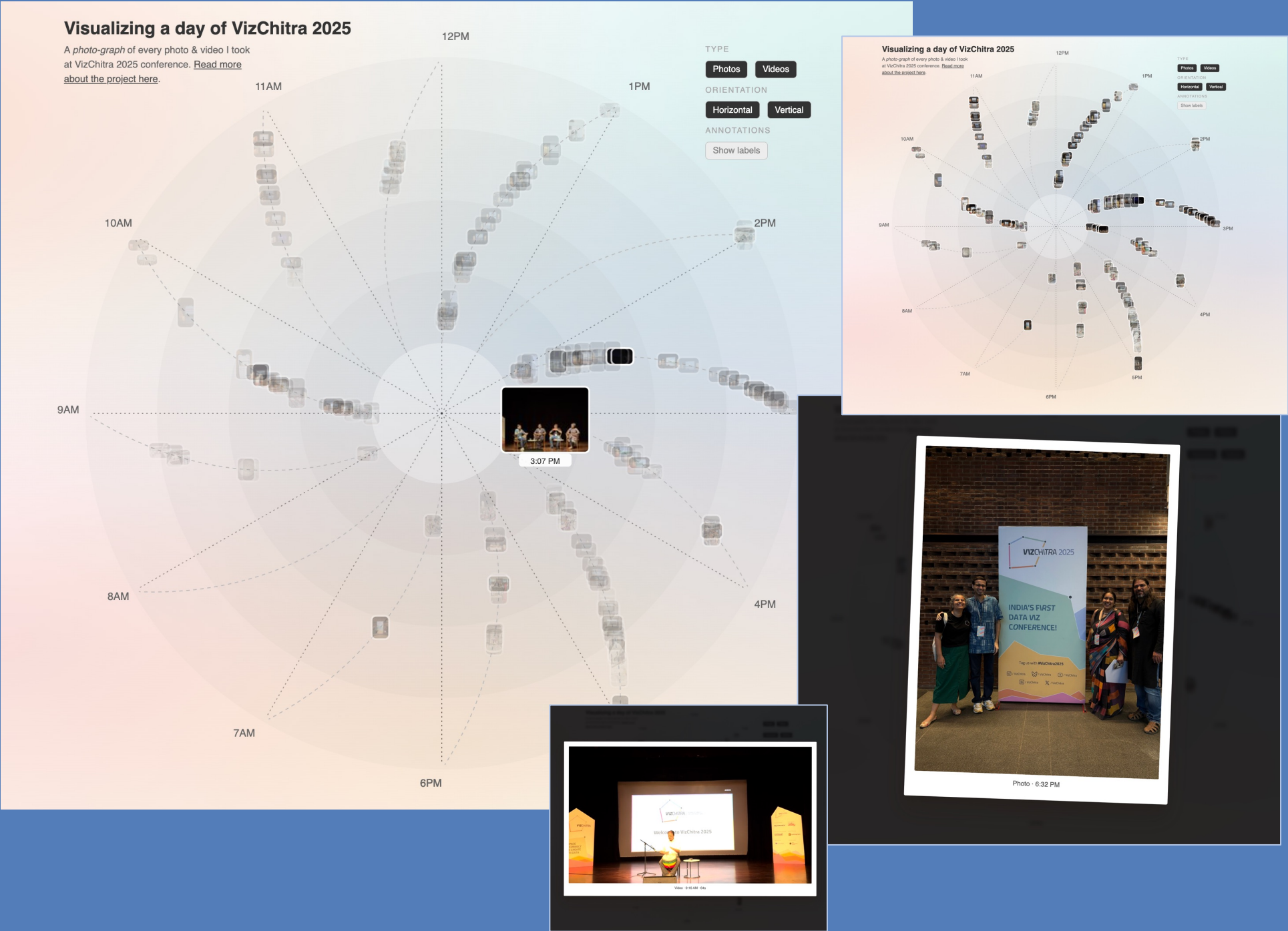
Flying above Fuji River - 2:57PM - 14 Nov, 2023

If time travel ever exists, I wish I could sit next to Hokusai on a flight, and watch him look at Mt. Fuji from this view.

Photo-graph:
26 shots of Mt. Fuji

What does my phone remember from an event I attended?

Photo-graph:
Visualizing a day of
VizChitra 2025



How does an AI visualize data on FIFA World Cup Group Stage?

My experiments using one-shot and skill orchestration to create interactive data viz.

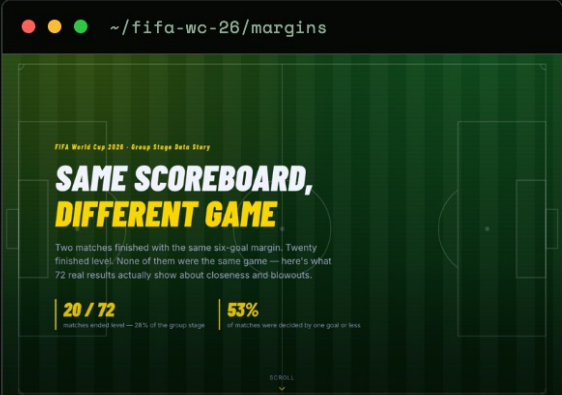
C:\FIFA-WC-26> RUN GALLERY.EXE

LOADING 6 DATA STORIES... OK

FIFA World Cup 2026 – Data Stories

A collection of data stories created using AI (Claude & Codex) about the FIFA World Cup 2026 group stages. Pick a terminal window below to open a story.

~/fifa-wc-26/margins



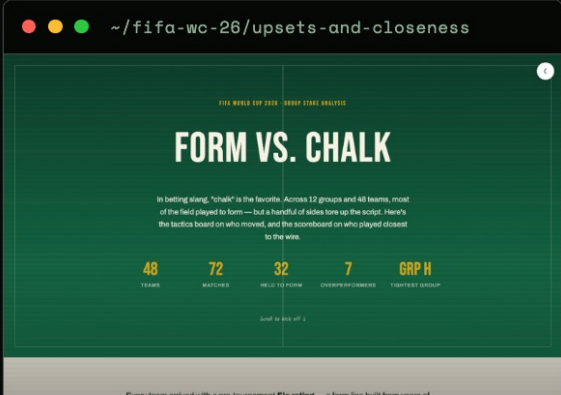
Same Scoreboard, Different Game

All 72 matches plotted on a margin spectrum, then zoomed into the 20 draws; insight: draws aren't equal — 53% of all matches were decided by one goal or less, and Algeria 3–3 Austria stands alone as the only high-scoring draw among 20.

Claude Code · /data-vizard

OPEN ▶

~/fifa-wc-26/upsets-and-closeness



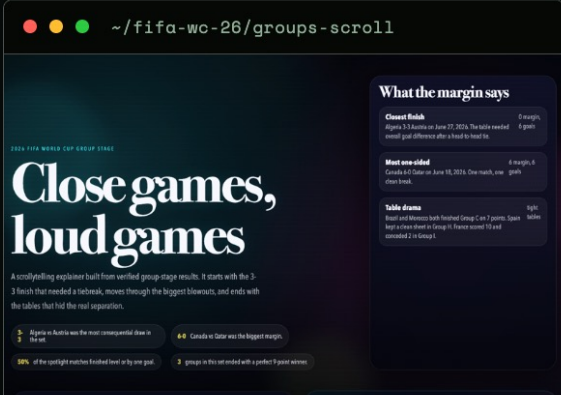
Form vs. Chalk

Plots all 48 teams' pre-tournament Elo expectation against actual finish alongside the same closeness spectrum; insight: most teams (33/48) finished exactly as expected, but standout upsets like South Africa and Cabo Verde jumped two spots while Group H was both the tightest group and home to one of the biggest overperformers.

Claude Code · /data-vizard

OPEN ▶

~/fifa-wc-26/groups-scroll



Close games, loud games

A scrollytelling walk through the closest, most one-sided, and most consequential group-stage matches; insight: closeness isn't just about who won — the "loudest" (highest-scoring) games weren't always the closest ones.

Codex · /data-vizard

OPEN ▶

The *AI wave* will transform how we design,
and what we will design.

To go beyond merely exhibiting AI output,
we must learn to swim in *data & technology*.

Resources shared in the class

References from Rasagy

- [16 variations of todo app UI](#)
- [Creative Coding JS Libraries](#)
- WIP collection of [resources for AI x Design](#)
- [Recreating paintings in P5.js](#)
- [AI traps to know about](#)

More inspiration

- [Emil Kowalski's animation skills](#)
- [Drawing data with kids by Gulrez Khan](#)
- [ColorBrewer2](#) for color palettes
- Archive of (AI) [Design Demo Nights](#) by Kenneth

Rasagy Sharma
(@rasagy.sharma / @data.n.coded)